Mathematical Equations of Neurotransmitter Systems

Chapter 4: Quantitative Models of Neurotransmitter Dynamics

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April 2, 2026

Abstract

This document presents the mathematical foundation of neurotransmitter systems. Every major neurotransmitter function described in Chapter 4 is converted into published computational and biochemical mathematical models. Equations cover synthesis kinetics, receptor binding/activation, signal transduction cascades, reuptake dynamics, and neuromodulatory systems.

Contents

1 Neurotransmitter Synthesis Kinetics **2**
- 1.1 Michaelis-Menten Enzyme Kinetics — General Synthesis Rate . . . 2
  - 1.1.1 Michaelis-Menten Equation . . . 2
- 1.2 Tyrosine Hydroxylase Feedback Inhibition . . . 2
  - 1.2.1 Competitive Inhibition by End Product . . . 3
  - 1.2.2 Phosphorylation Activation of TH . . . 3
- 1.3 Vesicular Neurotransmitter Loading . . . 3
  - 1.3.1 Vesicular Transport Rate . . . 3

2 Glutamate Receptor Models **3**
- 2.1 AMPA Receptor Kinetics — 4-State Markov Model . . . 3
  - 2.1.1 State Transitions . . . 4
  - 2.1.2 Conductance from Open State Fraction . . . 4
  - 2.1.3 Two-Exponential Approximation . . . 4
- 2.2 NMDA Receptor Model — Voltage-Dependent Mg^{2+} Block . . . 4
  - 2.2.1 NMDA Conductance with Mg^{2+} Block . . . 4
  - 2.2.2 Voltage-Dependent Mg^{2+} Blockade . . . 4

1 Neurotransmitter Synthesis Kinetics

1.1 Michaelis-Menten Enzyme Kinetics — General Synthesis Rate

What it models: Rate of neurotransmitter synthesis by biosynthetic enzymes (Tyrosine Hydroxylase for dopamine/norepinephrine; Glutamic Acid Decarboxylase for GABA; Tryptophan Hydroxylase for serotonin; Choline Acetyltransferase for ACh). All follow Michaelis-Menten kinetics.

Ref: Michaelis, L. & Menten, M.L. (1913). Die Kinetik der Invertinwirkung. *Biochemische Zeitschrift*, 49, 333–369.

1.1.1 Michaelis-Menten Equation

$$v = V_{\max} \frac{[S]}{K_m + [S]} \tag{1}$$

Variable	Definition	Typical Values
v	Reaction velocity (synthesis rate)	nM/min or μM/s
$V_{\max}$	Maximum velocity (saturated enzyme)	varies by enzyme
$[S]$	Substrate concentration	mM
K_m	Michaelis constant	μM–mM

Rate-Limiting Enzyme Parameters:

Enzyme	Neurotransmitter	Substrate	K_m
Tyrosine Hydroxylase (TH)	Dopamine/NE	L-Tyrosine	50–100 μM
Tryptophan Hydroxylase (TPH2)	Serotonin	L-Tryptophan	30–60 μM
Glutamic Acid Decarboxylase (GAD67)	GABA	Glutamate	0.6–2.0 mM
Choline Acetyltransferase (ChAT)	ACh	Choline	0.5–1.0 mM

Key insight: Since TH $K_m \approx 50$ μM and typical brain tyrosine is ˜100–150 μM, TH operates near half-saturation — synthesis rate is sensitive to tyrosine availability.

1.2 Tyrosine Hydroxylase Feedback Inhibition

What it models: TH (the rate-limiting enzyme for dopamine and norepinephrine synthesis) is inhibited by its own product — end-product inhibition provides homeostatic control.

Ref: Leviel, V. (2011). Dopamine release mediated by the dopamine transporter, facts and consequences. *Journal of Neurochemistry*, 118(4), 475–489.

1.2.1 Competitive Inhibition by End Product

$$v = V_{\max} \frac{[S]}{K_m \left(1 + \frac{[I]}{K_i}\right) + [S]} \tag{2}$$

Variable	Definition	Units
$[I]$	Concentration of inhibitor (dopamine or NE)	μM
K_i	Inhibition constant (affinity for inhibitor)	μM

1.2.2 Phosphorylation Activation of TH

$$V_{\text{max,phospho}} = V_{\text{max,basal}} \times k_{\text{phos}} \tag{3}$$

where $k_{\text{phos}} \approx$ 3–10-fold increase upon serine phosphorylation (Ser19, Ser31, Ser40).
Key insight: DA neurons regulate their own synthesis via two parallel mechanisms: (1) product inhibition by cytoplasmic dopamine, and (2) D2 autoreceptor signaling — inhibits TH phosphorylation — reduces synthesis.

1.3 Vesicular Neurotransmitter Loading

What it models: Packaging of neurotransmitter into synaptic vesicles by vesicular transporters (VGLUT for glutamate, VGAT for GABA, VMAT2 for monoamines, VAChT for ACh).

Ref: Parsons, S.M. (2000). Transport mechanisms in acetylcholine and monoamine storage. *FASEB Journal*, 14(15), 2423–2434.

1.3.1 Vesicular Transport Rate

$$\frac{d[NT]_v}{dt} = V_{\text{vesicle}} \frac{[NT]_{\text{cytosol}}}{K_T + [NT]_{\text{cytosol}}} - k_{\text{leak}}[NT]_v \tag{4}$$

Variable	Definition
$[NT]_v$	Vesicular neurotransmitter concentration
V_{vesicle}	Maximum vesicular uptake velocity
$[NT]_{\text{cytosol}}$	Cytosolic neurotransmitter concentration
K_T	Transporter affinity constant
k_{leak}	Passive leak rate

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2 Glutamate Receptor Models

2.1 AMPA Receptor Kinetics — 4-State Markov Model

What it models: Fast activation, rapid desensitization, and deactivation of AMPA receptors. This determines the time course of fast excitatory postsynaptic currents (EPSCs).

Ref: Destexhe, A., Mainen, Z.F. & Sejnowski, T.J. (1994). Synthesis of models for excitable membranes, synaptic transmission and neuromodulation. *Journal of Computational Neuroscience*, 1(3), 195–230.

2.1.1 State Transitions

$$C \underset{k_{-1}}{\overset{r_1[T]}{\rightleftharpoons}} C^* \underset{\beta}{\overset{\alpha}{\rightleftharpoons}} O \quad \text{(desensitization)} \tag{5}$$

2.1.2 Conductance from Open State Fraction

$$g_{\text{AMPA}}(t) = \bar{g}_{\text{AMPA}} \cdot O(t) \tag{6}$$

$$I_{\text{AMPA}} = g_{\text{AMPA}}(t)(V_m - E_{\text{AMPA}}) \tag{7}$$

2.1.3 Two-Exponential Approximation

$$g_{\text{AMPA}}(t) = \bar{g}_{\text{AMPA}} \left(e^{-t/\tau_{\text{decay}}} - e^{-t/\tau_{\text{rise}}}\right) \tag{8}$$

Parameter	Definition	Typical Values
r_1	Binding rate (glutamate)	1–10 $\mu\text{M}^{-1}\text{ms}^{-1}$
τ_{rise}	Rise time	0.2–1 ms
τ_{decay}	Decay time	2–10 ms
E_{AMPA}	Reversal potential	0 mV

2.2 NMDA Receptor Model — Voltage-Dependent Mg^{2+} Block

What it models: NMDA receptors as coincidence detectors requiring both glutamate binding AND postsynaptic depolarization. The Mg^{2+} block provides the voltage gate.

Ref: Jahr, C.E. & Stevens, C.F. (1990). Voltage dependence of NMDA-activated conductances. *Journal of Neuroscience*, 10(9), 3178–3182.

2.2.1 NMDA Conductance with Mg^{2+} Block

$$g_{\text{NMDA}}(V_m, t) = \bar{g}_{\text{NMDA}} \cdot s(t) \cdot B(V_m) \tag{9}$$

2.2.2 Voltage-Dependent Mg^{2+} Blockade

$$B(V_m) = \frac{1}{1 + \frac{[\text{Mg}^{2+}]_o}{K_{\text{Mg}}} \exp\left(-\frac{V_m}{\delta \cdot 25.7}\right)} \tag{10}$$

Jahr & Stevens (1990) parameters: $K_{\text{Mg}} = 3.57$ mM, $\delta = 0.89$

Parameter	Definition	Typical Values
$[\text{Mg}^{2+}]_o$	Extracellular Mg^{2+}	1–2 mM
τ_{NMDA}	NMDA decay time constant	50–300 ms
E_{NMDA}	Reversal potential	0 mV

Next: Chapter 5 — Neural Circuits and Networks